Project Report

On

<PROJECT TITLE IN CAPITAL>

Submitted to

Prof. Ram Meghe Institute of Technology and Research,

(An Autonomous Institute)

Affiliated to Sant Gadge Baba Amravati University, Amravati

for partial fulfillment of the requirement for the degree of

BACHELOR OF TECHNOLOGY

in

Artificial Intelligence and Data Science

Submitted By

<Students Name>

Under the Guidance of

<Guide Name>

Department of Artificial Intelligence and Data Science

Prof. Ram Meghe Institute of Technology & Research, Bandera - Amravati

2026-2027

Department of Artificial Intelligence and Data Science

Prof. Ram Meghe Institute of Technology & Research, Badnera - Amravati

(An Autonomous Institute)

2026 - 2027

CERTIFICATE

*This is to certify that the **Project (7SA08)** entitled*

<Project title>

is a bonafide work and it is submitted to the Prof. Ram Meghe Institute of Technology and Research (An Autonomous Institute), affiliated to

Sant Gadge Baba Amravati University, Amravati

By

<Students Name>

in the partial fulfillment of the requirement for the degree of ***Bachelor of Technology*** *in* ***Artificial Intelligence and Data Science****, during the academic year* ***2026-2027*** *under my guidance.*

<Guide Name>
Guide
Department of AI & DS
Prof. Ram Meghe Institute of Technology & Research, Bandera- Amravati

Dr. S. R. Gupta
Head,
Department of AI & DS
Prof. Ram Meghe Institute of Technology & Research, Bandera -Amravati

External Examiner

_____

ACKNOWLEDGEMENT

With immense pleasure and deep gratitude, we would like to express our sincere thanks to all those who have supported and guided us throughout the completion of this project. This work stands as a testament to the inspiration, motivation, and technical assistance provided by many individuals. Without their continuous support and encouragement, this project would not have been possible.

We are profoundly grateful to our respected guide, **Guide Name**, for their valuable guidance, constant supervision, constructive suggestions, and for providing the necessary resources required for the successful completion of this project. Their patience, encouragement, and insightful feedback greatly contributed to the quality of our work.

We would also like to extend our sincere appreciation to **Dr. S. R. Gupta**, Head of the Department of Artificial Intelligence and Data Science, for providing us with the opportunity and encouragement to undertake this project. We are equally thankful to all the faculty members of the Department of Artificial Intelligence and Data Science for their kind cooperation, continuous encouragement, and academic support.

We owe a special note of gratitude to the staff members of the Department of Artificial Intelligence and Data Science for their direct and indirect assistance throughout this project.

We also express our heartfelt thanks to our colleagues for their collaboration, teamwork, and valuable contributions in developing this project. Their support and cooperation played a significant role in its successful completion. We are also thankful to the author research work is referred and used for the project work.

Finally, we extend our deepest gratitude to our parents, friends, and well-wishers for their constant encouragement, moral support, and understanding. Above all, we thank the Almighty God for guiding us and blessing us with the strength and wisdom to accomplish this work.

<ADD Student Name>

1.

2.

3.

4.

Format for B.Tech Project Report

Cover Page	: **Black** Laminated paper with **Sliver** Print and **Plastic** transparent cover sheet / layer
Page size	: A4
Printing	: Both Side Printing is compulsory *(from Chapter 1 onwards)*
Margins	: Left: - 1.6 inch. Right, Top, Bottom: - 1.0 inch.
Header-Right	: Name of Project Title
Footer-Left	: PRMIT&R // Dept. AI&DS // 2026-27
Footer-Right	: Page No. (Font Arial, size-10)
Text font	: 12 point Times New Roman Regular
Spacing	: 1.5 (Line Spacing)
Chapter Name	: 14 points Times New Roman Bold
Heading/ Subheading	: 12 point Times New Roman Bold
Page Border	: apply only for front pages before introduction
Page Number	: apply from 1st Chapter, not before it
Report shall contain	:- TITLE page, CERTIFICATE page, Acknowledgement page, Abstract Table of Contents, List of Tables; List of figures; List of Screenshots (if any)

Chapter's Order: -

1. Introduction
2. Literature Review
3. System Architecture/Analysis
4. System Designs (Architecture/Block Diagram, DFD, Algo, etc.)
5. Implementation & Results
6. Conclusion (including future scope of the work)

   References (As per IEEE format)

   List of Published Paper in Conference, Journals (if any) and IPR filed.

   Resume of the Student in their Individual Copy and resumes of the all group members in Department copies.

Note: -

- *Total number of One + number of students in the group, copies of report should be printed*
- *As per requirement students may add more chapters with permission from Project Guide & Head.*

References (As per IEEE format)

Journal

Basic Format:

[1] A. A. Author, B. B. Author, and C. C. Author, "Title of the paper," *Abbreviated Journal Title*, vol. x, no. x, pp. xx–xx, Month Year, doi: xx.xxxx/xxxxx.

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[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *in Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol. (NAACL-HLT)*, Minneapolis, MN, USA, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

Books

Basic Format:

[1] J. K. Author, "Title of chapter in the book," in *Title of His Published Book, x*th ed. City of Publisher, Country if not India: Abbrev. of Publisher, year, ch. *x*, sec. *x*, pp. *xxx–xxx*.

Examples:

[1]. B. Klaus and P. Horn, *Robot Vision*. Cambridge, MA: MIT Press, 1986.

[2]. L. Stein, "Random patterns," in *Computers and You*, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.

[3]. R. L. Myer, "Parametric oscillators and nonlinear materials," in *Nonlinear Optics*, vol. 4, P. G. Harper and B. S.Wherret, Eds. San Francisco, CA: Academic, 1977, pp. 47-160.

[4]. M. Abramowitz and I. A. Stegun, Eds., *Handbook of Mathematical Functions* (Applied Mathematics Series 55).Washington, DC: NBS, 1964, pp. 32-33.

[5]. E. F. Moore, "Gedanken-experiments on sequential machines," in *Automata Studies* (Ann. of Mathematical Studies, no. 1), C. E. Shannon and J. McCarthy, Eds. Princeton, NJ: Princeton Univ. Press, 1965, pp. 129-153.

Handbooks

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Examples:

[1]. *Transmission Systems for Communications*, 3rd ed., Western Electric Co., Winston-Salem, NC, 1985, pp. 44–60.

[2]. *Motorola Semiconductor Data Manual*, Motorola Semiconductor Products Inc., Phoenix, AZ, 1989.

[3]. *RCA Receiving Tube Manual*, Radio Corp. of America, Electronic Components and Devices, Harrison, NJ, Tech. Ser. RC-23, 1992.

Reports

The general form for citing technical reports is to place the name and location of the company or institution after the author and title and to give the report number and date at the end of the reference.

Basic Format:

[1] J. K. Author, "Title of report," Abbrev. Name of Co., City of Co., Abbrev. State, Rep. *xxx*, year.

Examples:

[1]. E. E. Reber absorption in the earth's atmosphere," Aerospace Corp., Los Angeles, CA, Tech. Rep. TR-0200 (4230-46)-3, Nov. 1988.

[2]. J. H. Davis and J. R. Cogdell, "Calibration program for the 16-foot antenna," Elect. Eng. Res. Lab., Univ. Texas, Austin, Tech. Memo. NGL-006-69-3, Nov. 15, 1987.

[3]. R. E. Haskell and C. T. Case, "Transient signal propagation in lossless isotropic plasmas," USAF Cambridge Res. Labs., Cambridge, MA, Rep. ARCRL-66-234 (II), 1994, vol. 2.

Online Sources

FTP

Basic Format:

[1] J. K. Author. (year). *Title* (edition) [Type of medium]. Available FTP: Directory: File:

Example:

[2] R. J. Vidmar. (1994). *On the use of atmospheric plasmas as electromagnetic reflectors* [Online]. Available FTP: atmnext.usc.edu Directory: pub/etext/1994 File: atmosplasma.txt

WWW

Basic Format:

[1] J. K. Author. (year, month day). *Title* (edition) [Type of medium]. Available: http://www.(URL)

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[1] J. Jones. (1991, May 10). *Networks (2nd ed.)* [Online]. Available: http://www.atm.com

E-Mail

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Example:

[1] S. H. Gold. (1995, Oct. 10). *Inter-Network Talk* [Online]. Available e-mail: COMSERVE@RPIECS Message: Get NETWORK TALK

Patents, Standards, Theses, Unpublished

Patents

Basic Format:

[1] J. K. Author, "Title of patent," U.S. Patent *x xxx xxx,* Abbrev. Month, day, year.

Example:

[1] J. P. Wilkinson, "Nonlinear resonant circuit devices," U.S. Patent 3 624 125, July 16, 1990.

NOTE: Use "issued date" if several dates are given.

Theses (M.S.) and Dissertations (Ph.D.)

Basic Format:

[1].J. K. Author, "Title of thesis," M.S. thesis, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

[2].J. K. Author, "Title of dissertation," Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

Examples:

[1].J. O. Williams, "Narrow-band analyzer," Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993.

[2].N. Kawasaki, "Parametric study of thermal and chemical non equilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.

ABSTRACT

Abstract should be in 12 TNR ……

Table of Contents

Chapter No		Content / Title	Page No
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		List of Figures	*ii*
		List of Screenshots	*iii*
		Notations	*iv*
		Abbreviations	*v*
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	1.2	Motivation	3
	1.3	Problem Definition	4
	1.4	Project Goal and Objectives	6
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3.		**System Architecture / Analysis**	**35-86**
4.		**System Design**	**87-110**
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6.		**Conclusions and Future Scope**	
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List of Screenshot

Screenshot No.	Name of Screenshot	Page No.
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Guideline for preparation of project report for B.Tech.

1 ARRANGEMENT OF CONTENTS

The sequence in which the project report material should be arranged and bound should be as follows:

1. Cover Page & Title Page
2. Certificate Page
3. Acknowledgement
4. Abstract
5. Table of Contents
6. List of Tables
7. List of Figures
8. List of Screenshots [if any]
9. List of Notation /
10. List of Abbreviations
11. Chapters
12. Appendices [if any]
13. References
14. List of Publications
15. Resume of the Students

The table and figures shall be introduced at the appropriate places.

2 PAGE DIMENSION AND BINDING SPECIFICATIONS

The dimension of the project report should be in **A4** size, **TWO sides** printing.

Cover Page : White Laminated with plastic transparent sheet

Page size : A4

Printing : Both Side Printing is compulsory (from Chapter 1 onwards)

Margins : Left: - 1.6 inch. Right, Top, Bottom: - 1.0 inch.

Header-Right : Name of Project Title
Footer-Left : PRMIT&R // Dept. AI&DS // 2026-27
Footer-Right : Page No. (Font Arial, size-10)
Text font : 12-point Times New Roman Regular
Spacing : 1.5 (Line Spacing)
Chapter Name : 14 points Times New Roman Bold
Heading / Subheading : 12-point Times New Roman Bold
Page Border : apply only for front pages before introduction
Page Number : apply from 1st Chapter, not before it

3 PREPARATION FORMAT

3.1 Cover Page and Title Page

A specimen copy of the Cover page & Title page of the report is given in respective Template.

3.2 Abstract

Abstract should be maximum of two paragraphs of 200 to 300 Words and on single page of the project report typed in 1.5 line spacing, Font Style Times New Roman and Font Size 12.

3.4 Table of Contents

The table of contents should list all material following it as well as any material which precedes it. For numbering, **use roman lower-case number** from title page to list of abbreviation and normal number start from chapter 1 to the end of report as shown in template**.** A specimen copy of the Table of Contents of the project report is given in Template.

3.5 List of Tables

The list should use exactly the same captions as they appear above the tables in the text.

3.6 List of Figures

The list should use exactly the same captions as they appear below the figures in the text.

3.7 List of Symbols and Abbreviations

All the Standard symbols, abbreviations etc. used in the report should be explained under this head.

3.8 Chapters

The chapters may be broadly divided into 3 parts (i) Introductory chapter, (ii) Chapters developing the main theme of the project work (iii) and Conclusion. The main text will be divided into several chapters and each chapter may be further divided into several divisions and sub-divisions.

- Each chapter should be given an appropriate title.
- Tables and figures in a chapter should be placed in the immediate vicinity of the reference where they are cited.
- Footnotes should be used sparingly. They should be typed single space and placed directly underneath in the very same page, which refers to the material they annotate.

3.9 Appendices

Appendices are provided to give supplementary information, which is included in the main text may serve as a distraction and cloud the central theme.

- Appendices should be numbered using Arabic numerals, e.g. Appendix 1, Appendix 2, etc.
- Appendices, Tables and References appearing in appendices should be numbered and referred to as appropriate places just as in the case of chapters.

- Appendices shall carry the title of the work reported and the same title shall be made in the contents page also.

3.10 References

The listing of references should be typed (2 lines with 1.5 line spaces) below the heading "**References**" descending year wise in **single spacing** left – justified. The references should be formatted in IEEE Format. A typical illustrative list is given in UG Project Report Format Fil. The references citation should be made in the report wherever necessary; the example for citation of reference inside the report is [1], [1 – 5]; [1, 7, 9, 11].

References (TNR-14 bold)

References (As per IEEE format)

Journal

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Patents, Standards, Theses, Unpublished

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[2]. N. Kawasaki, "Parametric study of thermal and chemical non equilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.

.

3.11 Table and Figures

By the word Table, is meant tabulated numerical data in the body of the project report as well as in the appendices. All other non-verbal materials used in the body of the project work and appendices such as charts, graphs, maps, photographs and diagrams may be designated as figures. Caption of a table must be given only at the top of the table as Table X.Y (Caption/title). X will represent Chapter number whereas Y will represent the chronological number of that Table in the chapter. Use Times New Roman, Font 11, Bold, Centered. For figure, it must be only at the bottom as Fig. X.Y (caption/title). X will represent Chapter number whereas Y will represent the chronological number of that Table in the chapter Use Times New Roman, Font 11, Bold, Centered.

4. TYPING INSTRUCTIONS

The impression on the typed copies should be black in colour. **1.5 line spacing** should be used for typing the general text. The general text shall be typed in the Font style 'Times New Roman' and Font size 12 at 1.5 line spacing. Main body should be "Justified aligned", as given in Chapter Template.

Chapter Number and Name. – It should be Centre aligned. Font TNR-14, Bold, Uppercase as given in Template.

Sections / Subsections – Each Section/ subsection should start at 1.5 line spaces from last line of previous section. All Caption for section/ subsection should be left aligned; TNR-12 in Bold Uppercase, Bold Sentence case, Sentence case for section, subsection and sub-subsection as given in Chapter Template.

Paragraph – Each paragraph should start on new line with 0.5" indeed at first line only. In the paragraph setting line spacing before is 6 pt and after is 0 should be set with line spacing as 1.5 lines.

Note:

1) Chapter should not be separated by any blank page/ Transference.
2) Photograph should not be pasted directly (Color printout on A4 paper may be provided in appropriate manner at necessary location).
3) Student can use given Template for writing the text by overwriting it.
4) Start every chapter on right hand side page (Use new Template for each chapter).

RESUME

PERSONAL DETAILS	
NAME	
DOB	
ADDRESS	
MOBILE NO	
EMAIL_ID	
LINKEDIN ID	

Photo

EDUCATION DETAILS					
Name of Board			**Passing Year**		**% of Marks/ CGPA**
10th SSC	Maharashtra State Board				
12th HSC	Maharashtra State Board				
Diploma	Maharashtra State Board Of Technical Education				
Bachelor of Technology (B.Tech)		Passing Year & Month	Marks Obtain / Out of	% of Marks	Pointer
I st Year	I - SEM				
	II - SEM				
II nd Year	III - SEM				
	IV - SEM				
III rd Year	V - SEM				
	VI - SEM				
IV th Year	VII - SEM				
	VIII - SEM				

PLACEMENT DETAILS	
Campus Placement (If Any)	
(If Any) Name Of Company	

FUTUTRE PLANNING		
Higher Studies/ Job Preferences	Higher Studies	
	Job	
	Training	
	Business	

Place: Amravati
Date: / /2026

Signature
Name of Student

CHAPTER 1 (TNR 14 BOLD, UPPERCASE)

INTRODUCTION (TNR 14 BOLD, UPPERCASE)

1.1 INTRODUCTION (TNR 12 BOLD, UPPERCASE)

Piping systems are considered as the lifelines of industrial units such as process industries, fossil and fissile fuel power plants, as they comprise important components of these installations. It is crucial that the pipe networks remain functional in the event of strong motion earthquakes. Thus, it is of basic importance that the response behaviour of these systems under seismic excitations be well understood. To achieve this, significant efforts during the design phase are spent to ensure that the piping systems should behave as designed and the vibrations in the piping systems remain within permissible levels specified by design codes.

Generally, in normal operating conditions, piping systems are subjected to different type of normal loads such as dead load, pressure and load due to thermal expansion. However, during an earthquake, it will be subjected to seismic loads in addition to the normal loads. Piping systems are normally installed on the structures in which acceleration at the location of piping support is higher than the ground acceleration. Seismic loads generated on piping systems due to earthquakes can cause excessive vibrations, which can lead to high stresses resulting in damage or complete failure. Hence, the design of these piping systems is governed by two requirements. On the one hand, they need to be flexible enough to withstand normal thermal loading conditions and, on the other hand, they are to be stiff enough to limit the earthquake-induced displacements. A remedy to this conflict is achieved by supporting the piping systems and mountings (valves, traps, etc.) by seismic supports called snubbers. These seismic restraints allow the gradual movement of piping system for thermal loading and swings into action only during an occurrence of earthquake by restraining the piping systems.

The conventional design requires preventing the failure of piping systems by providing with the stronger structural elements to absorb and dissipate the transmitted earthquake energy elastically. Moreover, the piping systems with seismic supports like snubbers are associated with operational problems like oil leakage, inadvertent locking-

up and need frequent attention (Olson and Tang, 1988; Cloud *et al*., 1988). Undesirable load on piping systems due to malfunctioning of snubbers is another drawback associated with them (Jonczyk and Gruner, 1991).

1.2 SUPPLEMENTAL CONTROL DEVICES (TNR 12 BOLD, UPPERCASE)

Passive, active, semi-active and hybrid control devices also known as supplemental devices may

1.2.1 Passive Control Devices (TNR 12 Bold, Sentence case)

1.2.1.1 Viscous Damper (VD) (TNR 12, Sentence case)

1.2.1.1.1 Properties of VD (TNR 12, Sentence case, Italic)

1.2.1.2 Visco-elastic Damper (VED) (TNR 12, Sentence case)

1.2.2 Active Control Devices (TNR 12 Bold, Sentence case)

The active control systems are .....

1.2.3 Semi-active Control Devices (TNR 12 Bold, Sentence case)

1.2.4 Hybrid Control Devices (TNR 12 Bold, Sentence case)

A hybrid control system .....

1.3 BENEFIT OF SUPPLEMENTAL DEVICES (TNR 12 BOLD, UPPERCASE)

Comparative performance of these supplemental.....

1.4 NEED OF THE PRESENT STUDY (TNR 12 BOLD, UPPERCASE)

The research accomplishments, reported in the literature so far, deal with only the proposals for

1.5 OBJECTIVES OF THE PRESENT STUDY (TNR 12 BOLD, UPPERCASE)

The main objective of the present study is

1.6 ORGANIZATION OF THE REPORT (TNR 12 BOLD, UPPERCASE)

This report is divided in seven chapters, which is followed by appendix, references cited and publication from the present research work. A brief description of the contents of each chapter is presented here.

Chapter 1 covers a brief introduction, background leading to the